

Spreading, swirling and slope

Paper companion to the interactive page

Name: _____ Class: _____ Date: _____

The interactive page shows that a fixed source strength spread over a shell fixes the inverse-square law, that a fixed circulation round a loop fixes the $1/r$ law of a wire, and that either field can be read as the negative slope of a potential graph. This sheet asks you to draw, count and derive these results by hand. Use $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$, $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$ and $g = 9.81 \text{ N kg}^{-1}$ at the surface of the Earth.

1. A fixed amount over a growing shell

Every closed shell around a point source is crossed by the same total number of field lines, or by the same power, so the value per unit area depends only on the size of the shell. The area of a sphere is $4\pi r^2$ and the circumference of a circle is $2\pi r$.

A1. Complete the table. The relative value is the field or intensity at radius r divided by its value at $r = 1 \text{ m}$. **[3]**

r / m	3D point source		2D source (line or ripples)	
	area $4\pi r^2 / \text{m}^2$	relative value	length $2\pi r / \text{m}$	relative value
1				
2				
3				
4				

A2. A charge Q sits at the centre of a sphere of radius r . The total electric flux through the sphere is Q/ϵ_0 , and by symmetry the field has the same magnitude everywhere on the surface. Use these two statements to derive $E = kQ/r^2$. **[3]**

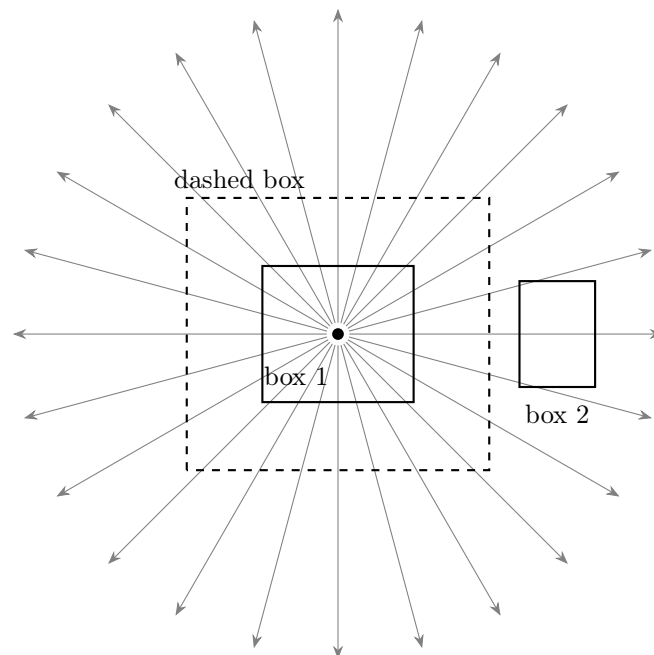
A3. A small lamp emits 100 W uniformly in all directions and none is absorbed. Calculate the intensity at (a) 2.0 m and (b) 6.0 m . **[3]**

- A4.** A very long tube lamp emits 40 W per metre of its length, and its light spreads out over cylinders. Calculate the intensity at (a) 0.50 m and (b) 2.0 m , and state the factor by which the intensity changes when the distance is quadrupled. [4]

- A5.** Calculate the gravitational field strength (a) at a height above the surface of the Earth equal to one Earth radius, and (b) at a height of two Earth radii. [3]

2. Cutting the flux with a probe box

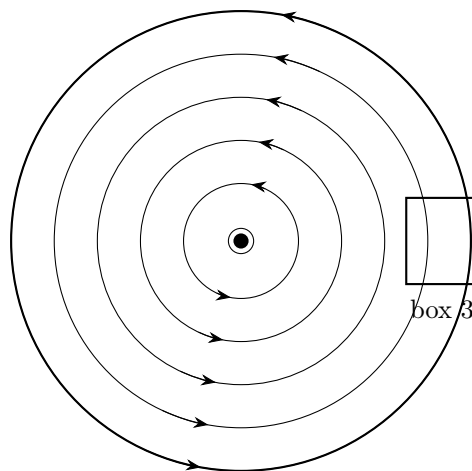
The diagram shows a point source with 24 field lines drawn at equal angles. Box 1 surrounds the source, box 2 sits away from it, and the dashed box surrounds the source at twice the edge of box 1.



- B1.** Count the field lines that leave box 1, and the lines that enter and the lines that leave box 2. [3]

- B2.** State how many lines leave the dashed box, and use your answers to explain why the net flux out of a box depends on whether the source is inside it and on nothing else about the box. [3]

Now consider the magnetic field of a long straight wire, with the current out of the page. The field lines are the closed circles drawn below, and box 3 lies beside the wire.



B3. Explain why the net magnetic flux out of box 3 is zero, and say whether the same conclusion holds for a box placed anywhere else in the field. [3]

B4. The current is 5.0 A. Calculate the sum of $B \Delta l$ round the circle of radius 2.0 cm, and state what the sum is for a circle of radius 4.0 cm. Explain what this constancy shows about the field. [4]

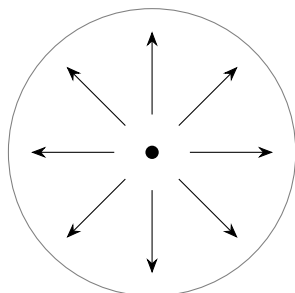
3. The wire and the quarter turn

C1. Use $\sum B \Delta l = \mu_0 I$ round a circle of radius r to derive an expression for the field of a long straight wire. [2]

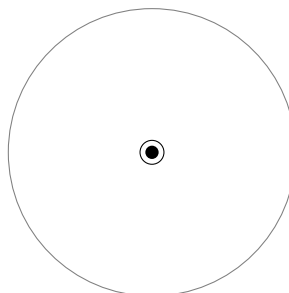
C2. A wire carries 8.0 A. Complete the table. [4]

r / cm	$B / \mu\text{T}$	$2\pi r / \text{m}$	$B \times 2\pi r / \mu\text{T m}$
1.0			
2.0			
4.0			

- C3.** A long charged rod has a field that points radially outward and falls as $1/r$, which is the two-dimensional version of the point charge. In the left circle below the radial field of the rod is drawn. In the right circle draw the field arrows of the current-carrying wire, which is the same pattern with each arrow turned through a quarter turn, and state the direction of the wire's field if the current points out of the page. **[3]**



line charge



wire, current out of the page

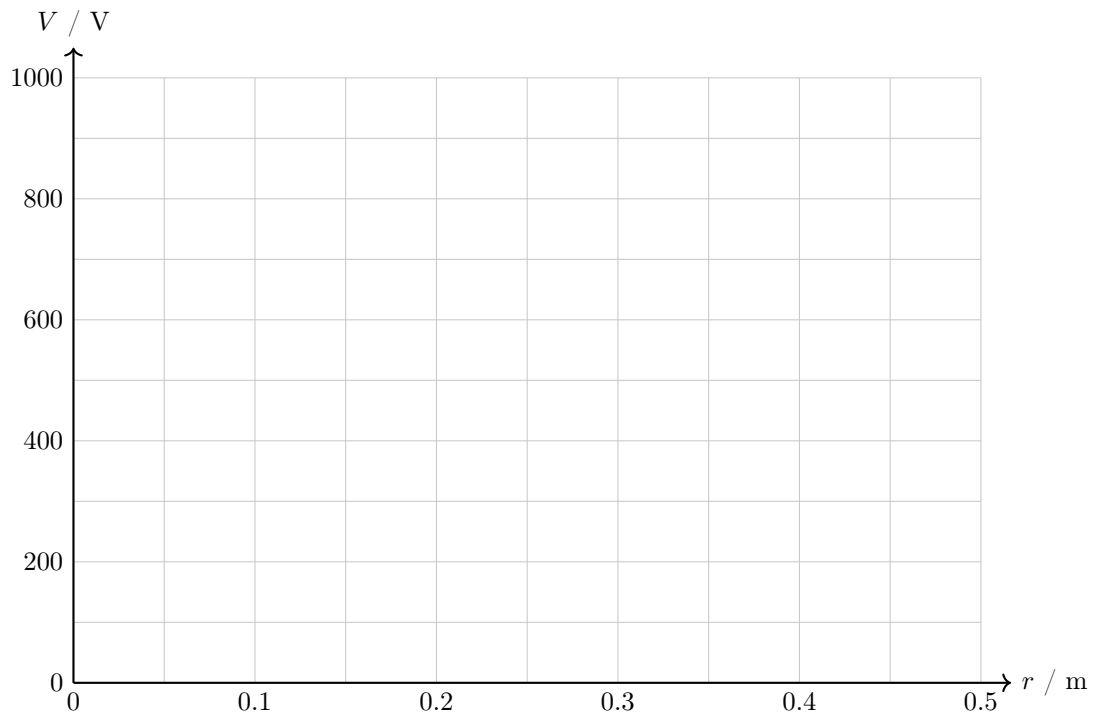
4. Reading the field from the slope of the potential

The radial component of the field is the negative of the gradient of the potential, $E = -\Delta V / \Delta r$ for electricity and $g = -\Delta V / \Delta r$ for gravity. A point charge of $Q = +5.0 \text{ nC}$ has potential $V = kQ/r$.

- D1.** Complete the table of potential values. **[2]**

r / m	0.05	0.10	0.20	0.30	0.40	0.50
V / V						

- D2.** Plot V against r on the grid and draw a smooth curve through your points. Then draw the tangent to the curve at $r = 0.10$ m. [3]



- D3.** Use the grid to find the gradient of the tangent at $r = 0.10$ m, and compare its magnitude with the value of kQ/r^2 at that point. State the direction of the field. [4]

- D4.** The potential of the Earth's field is $V = -GM/r$, which rises toward zero as r increases. State the sign of the gradient of the V - r graph and use it to state the direction of the gravitational field. [2]

- D5. Stretch.** A field that comes from a potential returns to the same value after a round trip, so its sum of $F \Delta l$ round any closed loop is zero. Use B2 and B4 to suggest why the magnetic field of a wire cannot be found as the gradient of a single-valued potential. [2]

5. Summary

Complete the table in your own words, so that each row states the shell or loop, what the source fixes, and the resulting law.

Field	Shell or loop	Fixed by the source	Law
Point charge, gravity, point lamp			
Line charge, tube lamp, ripples			
Straight wire (B)			
Any radial E or g field			